FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

OPEN TEST

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.
   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*
2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*
3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

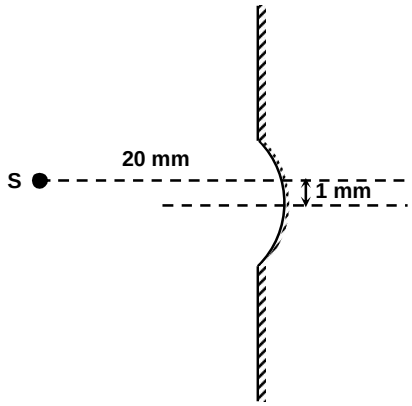
Physics

PART – I

SECTION – A
(One Options Correct Type)

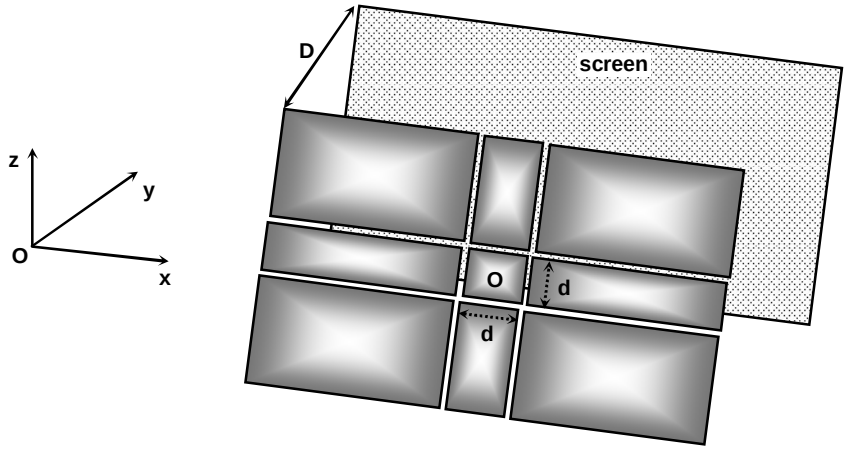
This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A practically very small source (S) of light is kept in front of a plane mirror. But due to imperfect manufacturing, there is a small spherical bulge (radius of curvature 80 mm). The situation is as shown in the figure. You can assume that principal axis of bulge is perpendicular to the plane mirror. Find the separation between the images formed.



(A) 10.025 mm
(B) 20.025 mm
(C) 40.025 mm
(D) 80.025 mm

2. Four slits are made in an infinitely large sheet as shown (O is the symmetrical point of the four slits):



Find the co-ordinates of the black spots formed on the screen on performing interference experiment with this arrangement (using light of wavelength λ), (m and n are integers):

(A) $\left[\frac{m\lambda D}{d}, D, \frac{n\lambda D}{d}\right]$

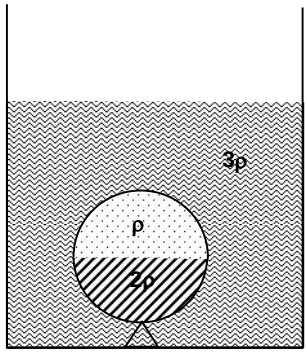
(B) $\left[\frac{(2m+1)\lambda D}{2d}, D, \frac{(2n+1)\lambda D}{2d}\right]$

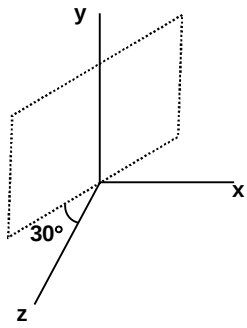
(C) $\left[\frac{(2m-1)\lambda D}{2d}, D, \frac{n\lambda D}{d}\right]$

(D) $\left[\frac{m\lambda D}{d}, D, \frac{(2n-1)\lambda D}{2d}\right]$

Space for Rough work

3. A solid sphere has density ρ in the top half and 2ρ in the bottom half. It is kept in a uniform fluid of density 3ρ with a hinge at the bottom (as shown) such that it can freely perform oscillations about any horizontal axis through the hinge (Radius of sphere is R). Find the period of small oscillations if all the points of the sphere are to oscillate in their respective planes parallel to the plane $x = -\sqrt{3}z$ (shown).





(A) $2\pi\sqrt{\frac{64R}{35g}}$

(B) $2\pi\sqrt{\frac{46R}{45g}}$

(C) $2\pi\sqrt{\frac{32R}{35g}}$

(D) $2\pi\sqrt{\frac{24R}{35g}}$

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. Two friends start cycling around a circular track (radius R) in the same sense starting from the same point with angular speed ω and 2ω. The slower one is continuously singing with his vocal cord frequency f_0 ($R\omega$ = 1% of v_{sound}). Then : (regarding the frequency of song heard by the faster one)

(A) Observed minimum frequency = $\frac{98}{99}f_0$.

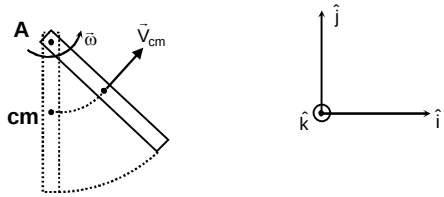
(B) Observed minimum frequency = $\frac{97}{99}f_0$

(C) Observed maximum frequency = $\frac{102}{101}f_0$

(D) The spectrum (of frequencies) repeats itself with a time period of $\frac{\pi}{\omega}$.

Space for Rough work

5. A uniform rod of length ℓ and mass m is rotating about a fixed axis perpendicular to the paper (xy-plane) and passing through its one of the end A. At the instant shown in the figure rod's angular velocity is $\vec{\omega}$ and velocity of the centre of mass of the rod is $\vec{v}_{cm}$. The moment of inertia of the rod about an axis passing through its centre of mass and parallel to $\hat{k}$ is I_{CM} and the moment of inertia of the rod about an axis passing through point A and parallel to $\hat{k}$ is I_A. $\vec{L}_A$ and k represent the angular momentum of the rod about point A and kinetic energy of rod respectively.



(A) $\vec{L}_A = I_A\vec{\omega} = \left(I_{cm} + \frac{m\ell^2}{4}\right)\vec{\omega},\ K = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

(B) $\vec{L}_A = I_A\vec{\omega} = m\vec{r}_{cm} \times \vec{v}_{cm} + I_{cm}\vec{\omega},\ K = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

(C) $\vec{L}_A = I_A\vec{\omega} = \left(I_{cm} + \frac{m\ell^2}{4}\right)\vec{\omega},\ K = \frac{1}{2}I_A\omega^2$

(D) $\vec{L}_A = I_A\vec{\omega} = m\vec{r}_{cm} \times \vec{v}_{cm} + I_{cm}\vec{\omega},\ K = \frac{1}{2}I_A\omega^2$

6. A uniform chain of mass m and length ℓ hangs from a hook in ceiling as shown in figure I. The bottom of link is now raised vertically with the help of upward external force F(y) very slowly as shown in figure II. Choose correct option(s)

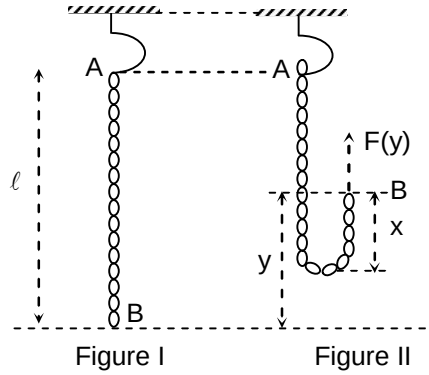


Figure I Figure II

(A) $F(y) = \frac{mgy}{4\ell}$

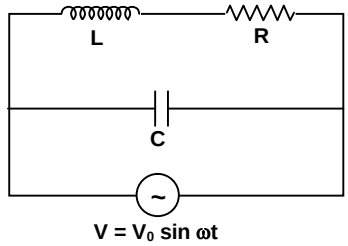
(B) $F(y) = \frac{mgy}{2\ell}$

(C) Work done by F(y) in raising end B to the position of end A is $\frac{mg\ell}{4}$

(D) Work done by F(y) in raising end B to the position of end A is $\frac{mg\ell}{8}$

Space for Rough work

7. Choose the correct options regarding to the following A.C. circuit:



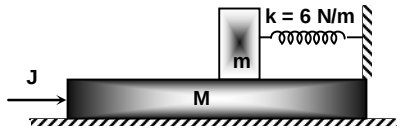
(A) Resonance occurs when $\omega = \frac{1}{\sqrt{LC}}$

(B) Resonance occurs when power factor is maximum.

(C) Resonance occurs when reactive power of circuit becomes equal to zero.

(D) Resonance occurs when $\omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2}$

8. A block of mass m = 1 kg is placed on a plank of mass M = 2 kg, which is placed on a horizontal frictionless surface. There is no friction between the block and the plank. Block and plank are connected by a spring of spring constant 6 N/m as shown in the figure. An impulse J = 5 newton-sec is applied on the plank. Possible magnitude of acceleration of the block is/are



(A) 1 m/s^2 (B) 3 m/s^2

(C) 5 m/s^2 (D) 7 m/s^2

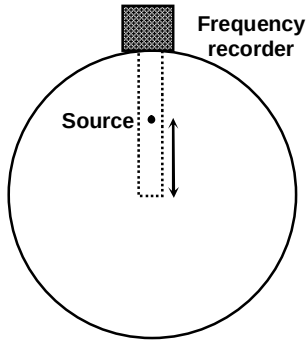
Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

A point source (frequency f_0) is dropped (by an observer) inside a very thin tunnel made from the surface of the earth (where acceleration due to gravity is g_0), with radius R, to the centre. The observer, at the surface, records the frequency received (f′) from the source. The situation is as shown. (speed of sound = v_0) :



9. If x is the distance of the source from the centre of earth and f′ is the frequency of the sound received which was emitted when the source was at x, then x is given by

(A) $x = R\sqrt{1-\frac{1}{2g_0R}v_0^2\left(\frac{f'}{f_0}-1\right)^2}$ (B) $x = R\sqrt{1-\frac{4}{g_0R}v_0^2\left(\frac{f_0}{f'}-1\right)^2}$

(C) $x = R\sqrt{1-\frac{2}{g_0R}v_0^2\left(\frac{f_0}{f'}-1\right)^2}$ (D) $x = R\sqrt{1-\frac{1}{g_0R}v_0^2\left(\frac{f'}{f_0}-1\right)^2}$

10. Find the location of the source after reaching which all signals emitted by the source are not received by the recorder. Ignore any reflections for the tunnel.

(A) $v_0\sqrt{\frac{R}{g}}$ (B) $\pi\sqrt{\frac{R}{g}}v_0$

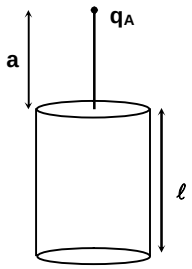
(C) $\sqrt{R^2-\frac{v_0^2R}{g}}$ (D) $\sqrt{\frac{R}{g}}v_0$

Space for Rough work

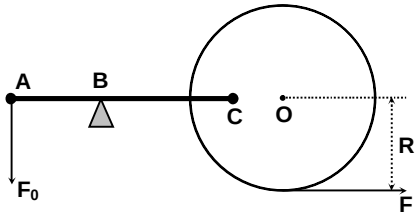
SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

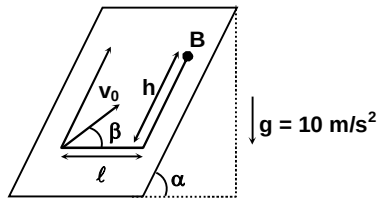
11. It is known that the electrically electrified bodies can attract the uncharged body. For example, the charged comb attracts the uncharged pieces of paper. One can estimate the force of interaction between charged body and uncharged body. For the estimation of force, consider an electrified body, localised at point A, having charge q and an uncharged body in form of conducting cylinder whose dimensions are considerably lower than distance from charge q ($a >> \ell$). The axis of cylinder is passing through point A. The volume of cylinder is V. If the force of interaction between them is $\frac{q^2}{\alpha\pi^2\varepsilon_0 a^5}V$. Find the value of α. (assume that no charge appears on the curved surface of cylinder)



12. A disc of radius 12 cm is free to rotate about point O, a rope is wound which is pulled with a force F = 10 N as shown in the figure. At point C, a light rod AC is attached to the disc, which rests on the fixed support at the point B. If force F_0 = 10k N is required to apply to the left end of the rod so that whole system remains in balance, find the value of k. Given BC = 30 cm, AB = 15 cm OC = 4 cm.

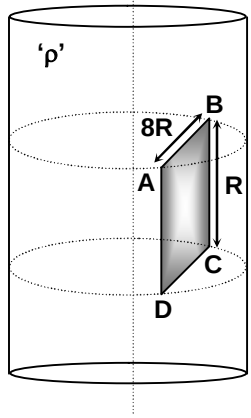


13. An inclined plane is located at angle α = 53° to the horizontal. There is a hole at point B in inclined plane as shown in the figure. A particle is projected along the plane with speed v_0 at an angle β = 37° to the horizontal in such a way so that it get into the hole. Neglect any type of friction. Find the speed v_0 (in m/s) if h = 1 m, ℓ = 8m and $\sqrt{5} \cong 2.25$.

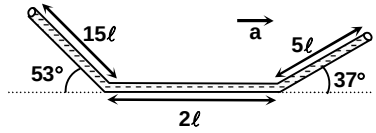


Space for Rough work

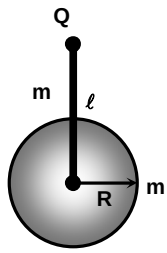
14. A non conducting solid cylinder of infinite length having charge density ρ and radius of cylinder is 5R. If the flux passing through the surface ABCD as shown in figure is $\frac{2\theta\rho R^3}{\varepsilon_0}$. Find the value of θ



15. In the given tube a liquid of density ρ is filled. If the acceleration of the tube towards right direction such that no liquid will falls out of the tube is $\frac{\eta g}{5}$. Find the value of η



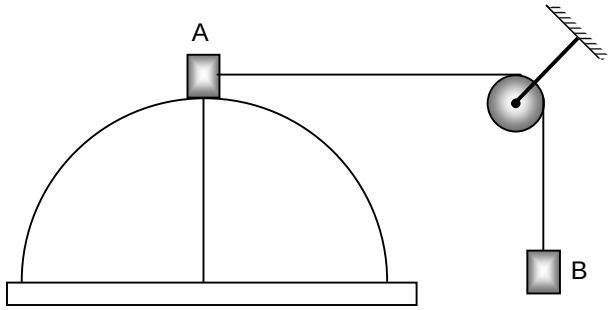
16. A meter stick connected with disc of same mass which is joined at one end of stick and disc can freely rotates about its joint. System swings in a vertical plane about a fixed horizontal axis passing through its one end, undergoes small oscillation of frequency 'f_0'. If disc is removed from the stick, then its new frequency of small oscillation is $\sqrt{\frac{13}{k}}f_0$. Find the value of k (take R = ℓ/2)



17. A trolley of mass 18 kg is pulled with a velocity which depends on time and given as v = 3t. Sand is falling vertically on the trolley at a rate of 18 kg/s. Assume sand sticks to the trolley. If rate of increase of Kinetic energy of the trolley at t = 2 sec is $\eta \times 324$. Find the value of η

Space for Rough work

18. A block 'A' of mass 1 kg connected with a hanging mass 'B' of 1 kg by an inextensible light string, is kept on a massive hemispherical surface of radius 2 m which is moving with an upward acceleration 4 m/s^2. The coefficient of friction between block and hemispherical surface is 0.5. If block 'A' is projected towards left with a velocity 1.5 m/s and the hemispherical surface move with a velocity 0.5 m/s towards right at the given position (shown in figure) then acceleration of block 'A' is $\eta\sqrt{17}$ m/s^2. Find the value of η



19. A hopper drops gravel at a rate of 15 kg/s onto a conveyor belt moving at a constant speed 4 m/s. If the output power must motor have in order to drive the conveyor belt is 48 k. Find the value of k

20. A parallel plate capacitor (square plates) has a capacitance of 1 mF. It is completely filled with a solid dielectric (constant = 5) which is performing very small oscillations (SHM, of amplitude 0.1% of plate dimension) with a period of 1 ms as shown: At all times, a battery of 10 V remains connected to the capacitor. If the amount of charge that crosses the battery in one cycle is 40 α. Find the value of α



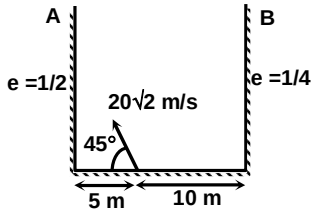
Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. A lamp post of length ℓ and cross sectional area S = 10 cm^2 is made of material with linear expansion coefficient $\alpha = 1.1 \times 10^{-5}\ K^{-1}$ and Young's modulus $Y = 2.0 \times 10^{11}\ N/m^2$. If temperature of lamp post is raised by ΔT = 10K, what mass (in kg) should be put on the lamp post, so that there is no change in the length of lamp post ($g = 10\ m/s^2$)

22. A ball is projected with velocity $20\sqrt{2}$ m/s at an angle of 45° with horizontal as shown in figure. The ball collide with the wall, the coefficient of restitution between the ball and wall A is $\frac{1}{2}$ and with the wall B is $\frac{1}{4}$, distance of point where ball collide with the ground from the projection point is



23. A cubical block of wood (density = 800 kg/m^3) and side length ℓ = 50 cm is floating in a rectangular tank partially filled with water (density = 1000 kg/m^3). Work done to press the block slowly so that it is just immersed in water is W Joules (Area of the bottom of tank is 10000 cm^2) (take $g = 10\ m/s^2$). Find the value of W

Space for Rough work

Chemistry

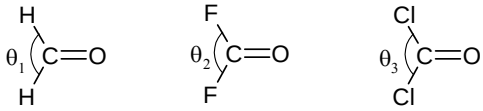
PART – II

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

24. Correct trend of first ionisation energy in group-13 is:
(A) B > Al > Ga > In > Tl
(B) B > Al > Ga > Tl > In
(C) B > Tl > Ga > Al > In
(D) B > Ga > Al > In > Tl

25. The self ionization constant for pure formic acid $K = \left[HCOOH_2^+\right]\left[HCOO^-\right]$ has been estimated as 10^{-6} at room temperature. The density of formic acid is 1.15 g/cm^3. What percentage of formic acid molecules convert to formate anion:
(A) 0.001%
(B) 0.002%
(C) 0.003%
(D) 0.004%

26. 
Decreasing order of shown angles will be:
(A) $\theta_1 > \theta_2 > \theta_3$
(B) $\theta_1 > \theta_3 > \theta_2$
(C) $\theta_3 > \theta_2 > \theta_1$
(D) $\theta_1 = \theta_2 = \theta_3$

Space for Rough work

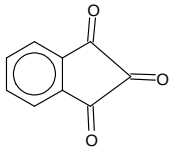
(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27. Between ClO_2 and Cl_2O:
(A) the Cl – O bond length would be larger in Cl_2O (B) the bond angle would be larger in Cl_2O
(C) ClO_2 is likely to be a coloured compound (D) Cl_2O is likely to be soluble in water

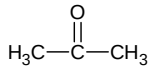
28. In the structure of **β,β′ -dichlorodiethyl sulphide,** more popularly known as *mustard gas:*
(A) all C-atoms are sp^3 hybridised
(B) S-atom is sp^2 hybridised
(C) there is a total of 2 lone pair of electrons on one molecule
(D) order of bond strength : C – C > C – Cl > C – S

29. Which of the following carbonyl compound(s) form (s) readily isolable hydrate?

(A) 

(B) $CCl_3 - CHO$

(C) 

(D) 

Space for Rough work

30. For the complex having the IUPAC name:
Dichloridotetrakis(ethylenediamine)chromium(III)tetrachloridopalladate(II), which of the following shall be true: (atomic nos. of Cr and Pd are 24 and 46 respectively)
(A) transition metal in the cation supercedes nearest noble gas configuration
(B) transition metal in the anion falls short of nearest noble gas configuration
(C) 1 molal aqueous solution of this complex will have the same freezing point as 1 molal NaCl solution in water
(D) 1 molar aqueous solution of this complex will be isotonic with 3 molar sucrose solution in water

31. Which of the given statement is correct:
(A) The C–Cl bond length in CH_3CH_2Cl is 1.76 Å and $CH=CH-Cl$ is 1.69 Å
(B) The C = C bond length in $CH_2 = CH_2$ is 1.34 Å and $CH_2 = CHCl$ is 1.38 Å
(C) The C – C bond length in $CH_2 = CH_2$, $CH_3CH_2 – Cl$ and $CH_2 = CHCl$ is 1.54 Å
(D) The dipole moment of CH_3CH_2Cl is 2.05 D and CH_2=CHCl is 1.44 D

Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 32 to 33

Redox reaction involves transfer of electrons between 2 chemical species. An **unbalanced** and **incomplete** example is shown below:

$$N_2 + H_3PO_2 \longrightarrow HN_3 + P_4O_6$$

32. If the above reaction were possible in either direction, maximum equivalent weight would belong to:
(A) N_2 (B) H_3PO_2
(C) HN_3 (D) P_4O_6

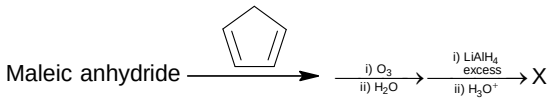
33. If the above reaction were unidirectional towards right, what fraction of P_4O_6 would be left unreacted if reaction were started with stoichiometric amount of reactants.
(A) $\frac{1}{2}$ (B) $\frac{3}{4}$
(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Space for Rough work

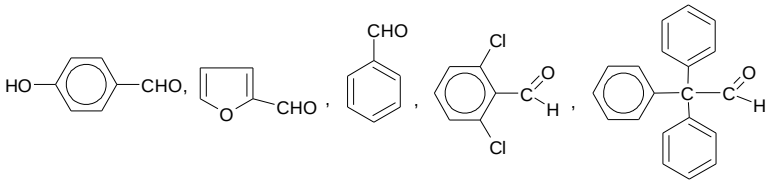
SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. The pH range of a basic indicator (InOH) is 3.4 to 4.6. For what minimum ratio of $\frac{[In^+]}{[InOH]}$ does the solution appear in the colour of In^+.

35. 

How many hydroxyl groups are present in X?

36. How many of the following compounds will give Cannizaro's reaction:


37. An aromatic compound P(C_8H_{10}) is heated with alkaline $KMnO_4$ to give a product which on heating with aqueous ammonia solution gives 'Q'. Compound 'Q' on heating with alkaline sodium hypochlorite gives a salt which on acidification gives product R ($C_7H_7O_2N$). Compound 'R' is treated with $NaNO_2$ & HCl and then with a base to produce a dipolar ion 'S' which on heating produces an intermediate (T) which dimerises to produce the final product 'U'.
What is the degree of unsaturation of the compound (U)?

38. The number of alkenes (open chain as well as cyclic) of six carbon giving same product with HBr in presence of peroxide and absence of peroxide.

39. 0.092 g of compound with molecular formula $C_3H_8O_3$ on reaction with an excess of CH_3MgI gives 67 mL of methane at STP. What is the number of active hydrogen atoms present in a molecule of the compound?

40. The isomerisation of cyclopropane (C_3H_6) to propene is believed to occur by the following mechanism:

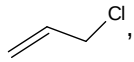
Step-1: $C_3H_6 + C_3H_6 \longrightarrow C_3H_6 + C_3H_6^*$

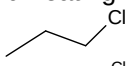
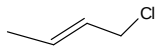
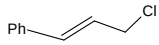
Step-2: $C_3H_6^* \longrightarrow CH_3 - CH = CH_2$

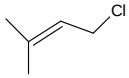
At high pressure, step-1 is bidirectional and fast whereas step-2 is unidirectional and slow. What is the order of the reaction?

Space for Rough work

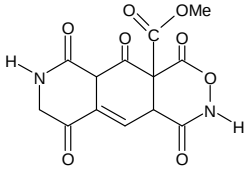
41. Ratio of frequency of revolution of electron in second excited state of He^+ and second state of H is $\frac{2^x}{3^y}$. What is the value of 'x + y'?

42. How many of the following undergo solvolysis by 50% aqueous ethanol, faster than  , on mild heating.

(a)  (b)  (c) 

(d)  (e)  (f) 

43. Consider the following compound (X):



If 0.5 moles of the compound (X) are hydrolysed in a alkaline medium and then heated. How many moles of CO_2 are produced?

Space for Rough work

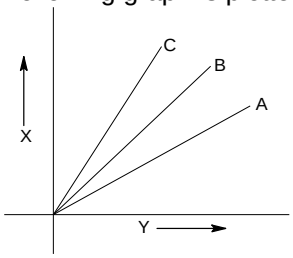
SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. Two inorganic compounds (A) and (B) are obtained by treating calcium orthophosphate with magnesium metal. (A) on hydrolysis produces a gas (C). The gas (C) reacts with $CuSO_4$ (aq) solution to give a black precipitate of compound (X). What is the molar mass of (X)? [Atomic mass of P = 31 g, Ca = 40 g, Mg = 24 g, O = 16 g, Cu = 63.55]

45. In an ABCABC packing of *Giant Buckies* (C – 240); *Bucky babies* – *Chotu* (C – 32) and *Motu* (C – 44) are doped in tetrahedral and octahedral voids respectively. In the unit cell thus created all balls along any one body diagonal are removed and *Bucky Onions* (C – 240 containing C – 60 suspended inside it, with their centres coinciding) are fitted at either extremes. The number of C atoms finally present in the unit cell are…….?

46. Following graph is plotted between characteristics 'X' and 'Y' for three species A, B and C:



Now consider the following statements, related to the above graph:

1. If 'X' is the velocity of electron and 'Y' is $\frac{1}{n}$ (where 'n' is n^{th} Bohr orbit of three hydrogen like species A, B and C), then radii of these species must increase in the order C < B < A.
2. If 'X' is the de-Broglie wavelength (λ) of three charged species A, B and C (having same amount of charge) and 'Y' is $\frac{1}{\sqrt{V}}$ (where 'V' is accelerating potential difference), then the mass of these species must increase in the order C < B < A.
3. If 'X' is the volume of same amount of three ideal gases A, B & C and 'Y' is the temperature in Kelvin, then the pressure of these gases must increase in the order C < B < A.
4. If 'X' is the pressure of the same amount of ideal gases A, B & C and 'Y' is $\frac{1}{V}$ (where V is the volume of these gases), then temperature of these gases must increase in the order: $T_C < T_B < T_A$.

What is the correct code for correct statements in increasing order of their codes given?

Space for Rough work

Mathematics PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

47. Solution of the differential equation $y^2\left(e^{x^3}+e^{y^3}\right)dy + x^2e^{x^3}\left(y^3-1\right)dx = 0$ is

(A) $e^{x^2}\left(y^3-1\right)+e^{y^3}=c$ (B) $e^{x^3}\left(y^3+1\right)-e^{y^3}=c$

(C) $e^{x^3}\left(y^3-1\right)+e^{y^3}=c$ (D) $e^{x^3}\left(y^2-1\right)+e^{y^2}=c$

(where c is arbitrary constant)

48. Number of positive integral solutions of $x_1x_2x_3x_4 = 3^{\alpha_1}\,5^{\beta_1}\,7^{\gamma_1}\,11^{\delta_1}$
(where $\alpha_1, \beta_1, \gamma_1, \delta_1$ are integers such that $0 \le \alpha_1 \le 1, 0 \le \beta_1 \le 1, 0 \le \gamma_1 \le 1, 0 \le \delta_1 \le 1$), is
(A) 625 (B) 580
(C) 550 (D) 650

49. In ΔABC, a is the arithmetic mean and b, c are two geometric means of two positive distinct numbers, then the value of $4^4\left(\frac{\sin^{12}B+\sin^{12}C}{\left(\sin2A+\sin2B+\sin2C\right)^4}\right)$ can be equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) 2 (D) 3

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. Let $A_1, A_2, A_3, \ldots, A_n$ be n points in plane such that no three of them are collinear and no two lines obtained by joining any two points are parallel. The number of new points of intersection obtained by intersection of lines joining any of these two points, is

(A) $3\cdot {}^nC_4$ (B) ${}^{{}^nC_2}C_2 - {}^{n-1}C_2 n$

(C) $\frac{{}^nC_2\cdot {}^{n-2}C_2}{2}$ (D) ${}^{{}^nC_2}C_2 - n(n-1)$

Space for rough work

51. Let f(x) be a non-constant, twice differentiable function defined on R such that y = f(x) is symmetric about line x = 1 and f(–1) = $f'\left(\frac{1}{4}\right) = f'\left(\frac{1}{2}\right) = 0$, then which of following is/are correct statement(s)?

(A) f″(x) = 0 has atleast four roots in the interval (0, 2)

(B) $\int_{-2}^{2}\left(x^5 + x^3\right) f\left(1+x\right)dx < \int_{0}^{2}\frac{1}{1+2^{f(x)}}dx$

(C) there exist at least one C $\in \left(\frac{3}{2}, \frac{7}{4}\right)$ such that f′(C) + Cf″(C) = 0

(D) for some C $\in \left(1, \frac{3}{2}\right)$, f′(C) = Cf″(C)

52. Let $I = \int_{0}^{\pi}\frac{\cos\theta}{10-6\cos\theta}d\theta$, then which of following is/are correct statement(s)

(A) I < 1 (B) I > 0

(C) I < 0 (D) I > 1

53. The line L is drawn through the point (1, 1) to intersect L_1 : x + 2y = 1 and L_2 : x + 2y = 3 at P and Q respectively. Line perpendicular to L from P intersects L_2 at R and line perpendicular to L from Q intersects L_1 and S. If the area of parallelogram PQRS is least, then which of the following is/are correct?

(A) the equation of line L can be y + 3x = 4 (B) the equation of line L can be 3y – x = 2

(C) the least area of PQRS is $\frac{8}{5}$ square unit (D) the equation of line L can be 3y – x = 4

54. Let ABCD be a rectangle inscribed in a circle whose one diameter is along line y = x. If coordinates of A and B are (0, 2) and (4, 3) respectively, then which of the following is/are correct?

(A) the coordinates of point D are $\left(\frac{1}{5}, \frac{6}{5}\right)$ (B) the coordinates of point D are (4, 6)

(C) the coordinates of C are $\left(\frac{21}{5}, \frac{11}{5}\right)$ (D) the coordinates of C are (6, 4)

Space for rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 to 56

Read the following write up carefully and answer the following questions:
Five persons shoot each other such that no one shoots himself, every one shoots exactly once and no shot is wasted, then

55. The number of ways such that all persons are hit is
(A) 60 (B) 44
(C) 120 (D) 36

56. The number of ways in which exactly one person is not hit is
(A) 420 (B) 180
(C) 210 (D) 540

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. If the number of onto functions from $\{1, 2, 3, 4\} \to \{2, 3, 4\}$ such that $f(i) \neq i$ for $i = 1, 2, 3, 4$ is k, then $\frac{k}{4}$ is equal to _____

58. Let the area enclosed by the curve $x^4 + y^4 = 2xy$ be A square units, then the value of [A] is equal to _____ (where [.] denotes the greatest integer function)

59. Let the probability of arranging 8 boys ($B_1, B_2, ..., B_8$) and 5 girls linearly such that B_1 and B_2 are not together and exactly four girls are together be p, then [99p] is equal to _____ (where [.] denotes the greatest integer function)

Space for rough work

60. If $3\sin\alpha\sin\beta + 4\cos\beta + 5\cos\alpha\sin\beta = 5\sqrt{2}\ \forall\ \alpha, \beta \in R$, then |adj adj A| is equal to _____

$\left\{\text{where } A = \begin{bmatrix} \sin\alpha & -\frac{1}{2} \\ \frac{1}{2} & \tan\beta \end{bmatrix}\right\}$

61. In ΔABC if $r_1 = 3r$, then the value of $9\left(\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{A}{2}\tan\frac{C}{2}\right)$ is equal to _____

(where r_1 and r are the ex-radius opposite A and the inradius of ΔABC respectively)

62. If α, β, γ are the roots of $x^3 + x^2 - 2x + 1 = 0$, then the value of $\sqrt{\Delta}$ is equal to _____

where $\Delta = \begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix}$

63. The value of $\dfrac{\sum_{r=1}^{k} 2^r\ {}^nC_r\ {}^{k-1}C_{r-1}}{\sum_{r=0}^{k} {}^nC_r\ {}^{n+k-r-1}C_{n-1}}$ (where $n \geq k$) is equal to _____

64. Let in ΔABC, $\angle A$ is $\frac{\pi}{2}$ and D is point on BC such that AD is altitude. If E, F and I are incentre of ΔABD, ΔADC and ΔABC respectively, then the value of $\frac{AI}{EF}$ is equal to _____

65. The value of $m^m - \sum_{r=1}^{m} \frac{(m+1)(r-1)m^{r-1}}{r\ {}^mC_r}$ is equal to _____

66. A variable chord of the hyperbola $\frac{x^2}{4} - \frac{y^2}{8} = 1$, subtends a right angle at the centre of the hyperbola. If this chord touches a fixed circle concentric with the hyperbola, then the square of radius of the fixed circle is equal to _____

Space for rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. P_1, P_2, P_3, ..., P_8 are 8 players of equal strength who play with each other in a knockout tournament. The probability that P_1 wins against P_2 and P_3 to reach the final and also wins the tournament is p, then 42p is equal to _____

68. Let $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left|e^{i\frac{2k\pi}{n}}-e^{i\frac{(k-1)2\pi}{n}}\right|$ = L, then $\frac{L}{10\pi}$ is equal to _____

69. The arithmetic mean of all integral values of real parameter α for which $\left|z-\left(\alpha^2-7\alpha+13+i\right)\right|=1$ and $\arg\ z\geq\frac{\pi}{2}$ is satisfied for at least one z is equal to _____

Space for rough work